

Relations

The word relation is used to indicate a relationship between two objects. There are many kinds of relationship in the world. We deal with relationship between students and teachers, an employee and his salary and so on. In mathematics, an example of relation is "less than" which is denoted by $<$, so that x is related to y if $x < y$. We note that a relation between two objects can be defined by listing the two objects as an ordered pair.

Relations on Sets

The most direct way to express a relationship between elements of two sets is to use ordered pair made up of two related elements. Let A and B be two ~~elements~~ sets as follows:

$$A = \{ \text{Calcutta, Patna, Lucknow, Chennai} \}$$

$$\text{and } B = \{ \text{West Bengal, Bihar, Uttar Pradesh, Tamil Nadu} \}$$

This is a relation "is a capital of" between the elements of Set A and Set B .

If R is used for the relation "is a capital of", then above information can be written as

Calcutta R West Bengal, Patna R Bihar,

Lucknow R Uttar Pradesh, Chennai R Tamil Nadu,

Omitting the letter R between the two elements we write the above information as an ordered pair, where

$$R = \{ (\text{Calcutta, West-Bengal}), (\text{Patna, Bihar}), (\text{Lucknow, Uttar Pradesh}), (\text{Chennai, Tamil Nadu}) \}$$

$$= \{ (x, y) : x \in A, y \in B, x R y \}$$

Thus the relation R is a subset of $A \times B$.

Definition $\rightarrow$ Let A and B be two sets. A relation from A to B is a subset of the cartesian product $A \times B$. Suppose R is a relation from A to B .

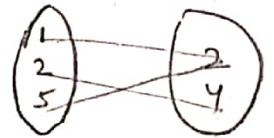
Then R is a set of ordered pairs (a, b) where $a \in A$ and $b \in B$. Every such ordered pair is written as $a R b$ and read as "a is related to b" by R .

Recall that the Cartesian Product $A \times B$ consists of all ordered pairs whose first element is in A and those whose second element is in B :

$$A \times B = \{ (x, y), x \in A \text{ and } y \in B \}$$

If $A = \{1, 2, 5\}$ and $B = \{2, 4\}$ then

$$A \times B = \{ (1, 2), (1, 4), (2, 2), (2, 4), (5, 2), (5, 4) \}$$



$$R = \{ (1, 2), (2, 4), (5, 2) \}$$

or $1R2, 2R4, 5R2$

→ Domain and Range → $R \subseteq A \times B$

The set $\{a \in A : (a, b) \in R \text{ for some } b \in B\}$ is called the domain of R and denoted by $\text{Dom}(R)$.

The set $\{b \in B : (a, b) \in R \text{ for some } a \in A\}$ is called the range of R and denoted by $\text{Ran}(R)$.

Ex → Let $A = \{2, 3, 4\}$ and $B = \{3, 4, 5\}$

List the elements of each relation R defined below and the domain and Range.

- Sol
- $a \in A, b \in B$ that is aRb if and only if $a < b$
 - $a \in A, b \in B, aRb$ iff a and b are both odd numbers

Soln $\Rightarrow$

$$R = \{ (2,3), (2,4), (2,5), (3,4), (3,5), (4,5) \}$$

$$\text{Dom}(R) = \{ 2, 3, 4 \}$$

$$\text{Ran}(R) = \{ 3, 4, 5 \}$$

Some operations on sets

If R and S denote two relations, then $R \cap S$ known as intersection of R and S , defines a relation such that

$$x(R \cap S)y = xRy \underset{\substack{\downarrow \\ \text{and}}}{\wedge} xSy$$

Similarly $R \cup S$, known as union of R and S such that

$$x(R \cup S)y = xRy \underset{\substack{\downarrow \\ \text{or}}}{\vee} xSy$$

Also

$$x(R - S)y = xRy \wedge x \notin y$$

$$\text{and } xR'y = xR'y \quad (R' \text{ is complement of } R)$$

Types of Relations in a Set

→ Inverse Relation → Let R be a relation from set A to set B . The inverse of R , denoted by R^{-1} is the relation from B to A which consists of ordered pairs which, when reversed, belong to R that is

$$\text{If } R = \{ (a, b) : (a, b) \in R \} \forall a \in A, b \in B \}$$

$$\text{then } R^{-1} = \{ (b, a) : (a, b) \in R \} \forall a \in A, b \in B \}$$

For ex → Let A be the set of all living people, Define relations B and C on A as follows

$$B = \{ (x, y) : x \text{ is a parent of } y \}$$

$$C = \{ (y, x) : y \text{ is a child of } x \}$$

then B and C are inverse of each other
i.e. $B = C^{-1}$ and $C = B^{-1}$.

(2) Identity Relation → A relation R in a set A is said to be identity relation, generally denoted by I_A , if

$$I_A = \{(x, x) \mid x \in A\}$$

For ex $\rightarrow$ If $A = \{1, 2, 3\}$

then $I_A = \{(1, 1), (2, 2), (3, 3)\}$ is an identity relation of A .

Properties of Relations $\rightarrow$

(iii) Empty Relation $\rightarrow$ If no element of A is in relation with any element of B , then the relation $R = \text{a Null set} = \phi = \{\}$ = an empty set. It is known as void or empty relation.

(iv) Universal Relation $\rightarrow$ Let A be a non-empty set and $R: A \times A \rightarrow A$ such that $\text{Domain } R = \text{Range } R = A$. The Relation R is known as universal relation.

Ex ① Let $A = \{2, 5, 7\}$ and R be a relation defined as $a R b$ ($a \neq b$) if and only if a divides b . Then we observe that $R = \phi$ is an empty relation.

Ex → ② Let $A = \{5, 6\}$ then $R = A \times A$

$= \{(5, 5), (5, 6), (6, 5), (6, 6)\}$ is a Universal relation.

Domain of $R = \text{Range of } R = \{5, 6\} = A$

Properties of Relations

A relation R on a set A satisfies certain properties.

These properties are defined as follows

(i) Reflexive Relation → A relation R on a set A is reflexive if $aRa \forall a \in A$, i.e. if $(a, a) \in R$ for every $a \in A$. This means that each element a of A is related to itself.

For ex → (i) $R_1 = \{(1, 1), (2, 2), (2, 3), (3, 3)\}$ be a relation on $A = \{1, 2, 3\}$, then R_1 is reflexive

(ii) $R_2 = \{(1, 1), (1, 2), (2, 3), (3, 3)\}$ be a relation on $A = \{1, 2, 3\}$, then R_2 is not reflexive relation since $(2, 2) \notin R_2$.

(ii) Irrreflexive relation $\rightarrow$ A relation R on a set A is irreflexive if, for every $a \in A$ $(a, a) \notin R$.

For ex $\rightarrow$

(a) The relation $R_1 = \{(1, 2), (1, 3), (2, 1), (2, 3)\}$ on $A = \{1, 2, 3\}$ is irreflexive relation since $(x, x) \notin R_1$ for every $x \in R_1$.

Non-reflexive relation $\rightarrow$ A relation R on a set A is non-reflexive if R is neither reflexive nor irreflexive, if $a R a$ is true for some 'a' and false for others. For example

$R = \{(1, 2), (2, 3), (2, 2), (3, 1)\}$ on $A = \{1, 2, 3\}$ is a non-reflexive relation since $2 R 2$ is true but $1 R 1$ and $3 R 3$ are false.

Symmetric Relation $\rightarrow$ A relation R on a set A is symmetric if whenever $(a, b) \in R$ then $(b, a) \in R$. i.e. if $a R b = b R a$.

Forex → $R_1 = \{ (1,1), (1,2), (1,3), (2,2), (2,1), (3,1) \}$ on $A = \{1,2,3\}$ ⑤
is a symmetric relation.

Asymmetric Relation → A relation R on a set A is
asymmetric if $(a,b) \in R \Rightarrow (b,a) \notin R$ i.e. if
 $aRb \neq bRa$ aRb $\nRightarrow bRa$ aRb $\nRightarrow bRa$

The relation $R_1 = \{ (1,1), (1,2), (1,3), (2,2), (2,1) \}$ on $A = \{1,2,3\}$
is an asymmetric relation. as $1R3 \neq 3R1$

Transitive Relation $\rightarrow$ A relation R on a set A is transitive if whenever $(a, b) \in R$, $(b, c) \in R$
 $\Rightarrow (a, c) \in R$. i.e.

$$a R b \text{ and } b R c \Rightarrow a R c.$$

or $UND \neq DND \rightarrow \dots$

Equivalence Relation $\rightarrow$ or RST Relative

A Relation R on a set A is called an equivalence relation or RST relative if it is reflexive, symmetric and transitive relation,

Thus, R is an equivalence relation on A if it has the following three conditions

- (i) $(a, a) \in R \quad \forall a \in A$ (Reflexive)
- (ii) $(a, b) \in R \Rightarrow (b, a) \in R, \quad \forall a, b \in A$ (Symmetric)
- (iii) $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R \quad \forall a, b, c \in R$
(Transitive)

(6)

Ex $\rightarrow$ 1) If R be a relation in the set of integers $\mathbb{Z}$ defined by

$$R = \{ (x, y) \mid x \in \mathbb{Z}, y \in \mathbb{Z}, (x-y) \text{ is divisible by } 6 \}$$

Prove that R is an equivalence relation

Solⁿ $\rightarrow$ Let $x \in \mathbb{Z}$ Then $x - x = 0$

and 0 is divisible by 6.

Therefore $x R x$ for all $x \in \mathbb{Z}$

Hence R is reflective (reflexive)

Again $x R y \Rightarrow (x - y)$ is divisible by 6

18 is divisible by 6
~~then~~ -18 is divisible by 6
 $18 \in \mathbb{Z}, -18 \in \mathbb{Z}$

$\Rightarrow -(x - y)$ is divisible by 6

$\Rightarrow (y - x)$ is divisible by 6

$\Rightarrow y R x$

Hence R is symmetric

$x R y$ and $y R z \Rightarrow (x - y)$ is divisible by 6 and

18 is divisible by 6

and 6 is divisible by 6

$(y - z)$ is divisible by 6

then $(18 + 6)$ is divisible by 6 $\Rightarrow [(x - y) + (y - z)]$ is divisible by 6

$\Rightarrow (x - z)$ is divisible by 6

$\Rightarrow x R z$

Hence, R is transitive

Thus R is an equivalence relation

Ex $\rightarrow$ Prove that in the set $A = \{1, 2, 3\}$ the relation
 $R = \{(1,1), (2,2), (3,3), (1,2), (2,1), (2,3), (3,2), (3,1), (1,3)\}$
is an equivalence relation

Soln

(i) Reflexive

Since $(1,1), (2,2), (3,3) \in R \Rightarrow aRa, \forall a \in A$

$\therefore R$ is reflexive

(ii) Symmetric

We have $(1,2)$ and $(2,1) \in R$

$(2,3)$ and $(3,2) \in R$

$(1,3)$ and $(3,1) \in R$.

$\therefore R$ is symmetric

(iii) Transitive

We have

when $(1,2), (2,3) \in R \Rightarrow (1,3) \in R$

$(2,3), (3,1) \in R \Rightarrow (2,1) \in R$

$(3,1), (1,2) \in R \Rightarrow (3,2) \in R$

$\therefore R$ is transitive

Thus R is an equivalence relation

Ex $\rightarrow$ If R is an equivalence relation of set A then R^{-1} is also an equivalence relation on A (7)

Soln

Since R is a relation on A

$$\Rightarrow R \subset A \times A$$

$$\Rightarrow R^{-1} \subset A \times A$$

$\Rightarrow R^{-1}$ is also a relation in A .

Now we have only to prove that R^{-1} is an equivalence relation.

i) Reflexive : Let $a \in A$, then

$$a \in A \Rightarrow (a, a) \in R \quad [\because R \text{ is reflexive}]$$

$$\Rightarrow (a, a) \in R^{-1}, \forall (a, a) \in A \times A$$

$\therefore R^{-1}$ is reflexive

(ii) Symmetric : Let $(a, b) \in R^{-1} \Rightarrow (b, a) \in R^{-1}$

$$(a, b) \in R^{-1} \Rightarrow (b, a) \in R$$

$$\Rightarrow (a, b) \in R \quad [\because R \text{ is symmetric}]$$

$$\Rightarrow (b, a) \in R^{-1}$$

$$\frac{a R^{-1} b = b R^{-1} a}{\therefore R^{-1} \text{ is symmetric}}$$

(iii) Transitive : Let $(a, b) \in \bar{R}'$ and $(b, c) \in \bar{R}'$,

Then $(a, b) \in \bar{R}'$ and $(b, c) \in \bar{R}'$

$\Rightarrow (b, a) \in R$ and $(c, b) \in R$

$\Rightarrow (c, b) \in R$ and $(b, a) \in R$

$\Rightarrow (c, a) \in R$ $\xrightarrow{[R \text{ is transitive}]}$

$\Rightarrow (a, c) \in \bar{R}'$ $\quad [\because R \text{ is transitive}]$

Hence $\bar{R}'$ is also an equivalence relation on A .

$(a, b) \in \bar{R}'$ and $(b, c) \in \bar{R}' \Rightarrow (a, c) \in \bar{R}'$

→ If R and S be two equivalence relations in a set A , then prove that $R \cap S$ is also an equivalence relation in A .

Soln R and S are relations in A

$$\Rightarrow R \subseteq A \times A \text{ and } S \subseteq A \times A$$

$$\Rightarrow R \cap S \subseteq A \times A$$

$\Rightarrow R \cap S$ is also a relation in A

Now we shall prove that it is an equivalence relation in A .

(i) Reflexive $\rightarrow$ Let $a \in A$, then

$$a \in A \Rightarrow (a, a) \in R \text{ and } (a, a) \in S \quad [\because R \text{ and } S \text{ are reflexive}]$$

$$\Rightarrow (a, a) \in R \cap S, \forall a \in A$$

$\therefore R \cap S$ is Reflexive

(ii) Symmetric $\rightarrow$ If $(a, b) \in R \cap S$ then $(b, a) \in R \cap S$

$$\bullet \text{ Now } (a, b) \in R \cap S \Rightarrow (a, b) \in R \text{ and } (a, b) \in S$$

$$\Rightarrow (b, a) \in R \text{ and } (b, a) \in S \quad (\because R \text{ \& } S \text{ are symmetric})$$

$$\Rightarrow (b, a) \in R \cap S$$

$\therefore R \cap S$ is symmetric

(iii) Transitive $\rightarrow$ If $(a, b) \in R \cap S$, $(b, c) \in R \cap S \Rightarrow (a, c) \in R \cap S$

$$\text{then } (a, b) \in R \cap S \Rightarrow (a, b) \in R \text{ and } (a, b) \in S$$

$$(b, c) \in R \cap S \Rightarrow (b, c) \in R \text{ and } (b, c) \in S$$

$$A = \{1, 2, 3, 4\}$$

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (3,4), (4,3)\}$$

$$S = \{(1,1), (2,2), (3,3), (4,4), (2,3), (3,2), (2,4), (4,2)\}$$

But R and S are transitive so

$$\left. \begin{array}{l} (a,b) \in R \text{ and } (b,c) \in R \Rightarrow (a,c) \in R \\ (a,b) \in S \text{ and } (b,c) \in S \Rightarrow (a,c) \in S \end{array} \right\} \Rightarrow (a,c) \in R \cap S$$

$\therefore R \cap S$ is transitive

Hence $R \cap S$ is an equivalence relation in A .

* Union of two equivalence relation not necessarily an equivalence relation